GOVERNMENT OF MAHARASHTRA · PUBLIC HEALTH DEPARTMENT

# NalamMesh

## Engineering Handover & Project Context

An offline-first public health platform for rural Maharashtra, modelled on Gadchiroli district. This document is everything a developer joining the project needs in order to work on it safely.

| | |
|---|---|
| **Problem Statement** | SIH 2026 · PS 26133 — rural healthcare access and continuity |
| **Repository** | github.com/TharunS2006/HealthcareAI — working branch **new** |
| **Stack** | Next.js 15 (App Router, static export) · React 19 · TypeScript · Zustand · IndexedDB · Socket.io relay · FastAPI district service |
| **Scale** | 45 routes · 13 on-device stores · 6 staff roles · 7 modelled facilities · 37 verify suites and 3 browser checks |
| **Status** | Software ready for a supervised pilot; approvals still due before real patients — see docs/PRODUCTION_READINESS.md |
| **Languages** | English, Hindi and Marathi — every citizen-facing screen is trilingual |

**Read this section first: the one rule that matters**

This is a medical application. **No figure shown to a user may be invented.** Every number must come from a real record, a real computation, or be replaced with an explicit "no data" message. Several rounds of work on this codebase have been spent removing fabricated values; do not reintroduce them. Section 8 explains this in full.

# 1. What the project is

NalamMesh is a public health platform built for the working conditions of Gadchiroli — a heavily forested district in eastern Maharashtra where connectivity is unreliable and the nearest specialist may be hours away.

## The three problems it addresses

| Problem | What the software does about it |
|---|---|
| **Connectivity fails. Facilities can be cut off for days.** | The device is the source of truth. All reads and writes hit local IndexedDB first; syncing to other facilities happens later, if a connection appears. |
| **Records don't travel. Paper registers stay at the facility.** | A patient's triage result, vitals and history move with the referral, so the receiving tier is not starting blind. |
| **The first contact is rarely a doctor — usually an ASHA or ANM.** | An on-device triage model scores urgency, with hard clinical rules layered above it that can only escalate, never downgrade. |

## The four-tier care network

Indian rural public health is organised in tiers. The whole data model follows this hierarchy — referrals move *up* the tiers, and a facility's tier determines which services and entitlements apply there.

| Tier | Facility | Modelled | Role |
|---|---|---|---|
| **SC** | Sub-Centre | 2 | First contact; ANM / ASHA staffed |
| **PHC** | Primary Health Centre | 2 | Medical Officer, basic laboratory |
| **CHC** | Community Health Centre | 1 | Specialists, emergency obstetric care |
| **SDH** | Sub-District Hospital | 1 | Surgery, inpatient beds |
| **DH** | District Hospital | 1 | Apex referral destination |

Facility records live in `lib/data/facilities.ts` — real names, coordinates, bed capacity, staffing and services for seven Gadchiroli facilities.

# 2. Architecture in one page

The single most important architectural fact: **the device is the source of truth.** The app is a statically exported Next.js site whose data layer is the browser's own IndexedDB. Two servers exist — the mesh relay and the district record service — but they are couriers and oversight: care continues when either is down. Understand that and most of the design follows.

## Request and data flow

```
UI component -> Zustand store -> lib/db.ts -> IndexedDB (on device)
                          |
                       mesh relay (:3001)  -> sign-in, staff, referrals
                       outbox -> district service (:8000) -> pre-arrival board
```

## Why each piece is there

| Piece | Why |
|---|---|
| **Static export** `output: 'export'` | No server to be unreachable. The whole app is files that can be served from anywhere, cached by a service worker, or wrapped in the Android build. |
| **IndexedDB via** `idb` | Structured, indexed, durable local storage. Survives refresh and restart. Schema is versioned — currently **v6** — with guarded upgrade handlers, checked by `verify:db-upgrade`. |
| **Zustand stores** | Thin layer between UI and the database. Writes are optimistic and roll back if the persist fails, so a clinical action never silently disappears. |
| **Service worker** `public/sw.js` | Hand-written, no PWA package. Pre-caches 32 routes and every file the build produced — the list is stamped into it after each build — so an installed app opens with no network. Currently **v4.1.0** plus the build id. |
| **TensorFlow.js** | Runs the triage model on the device. No inference request ever leaves the handset, so triage behaves identically offline. |
| **Socket.io mesh relay** | Checks PINs and signs session tokens, holds the staff directory, and relays referrals between facilities when any link exists. In production it keeps its state in a file and will not start without one. Work continues on the device without it. |
| **District record service** | FastAPI. Receives every patient record and referral from the device outbox so a receiving facility sees who is coming; serves the Data Inspector and the access log. Identified records: token-checked, access-logged, never cached. |
| **Capacitor** | Wraps the static export as an Android app for field devices. |

# 3. Where things live

Spend ten minutes here before writing any code.

| Path | Contents |
|---|---|
| **app/** | Next.js App Router pages — 45 routes. Citizen-facing at the root (`/opd`, `/queue`), staff screens under `/staff`. |
| **lib/db.ts** | **The heart of the project.** IndexedDB schema, every read and write, the audit trail. Read this file first. |
| **lib/triage/model.ts** | Triage engine: TensorFlow.js model plus the clinical override rules. See section 5. |
| **lib/data/** | Seed and reference data: facilities, hospitals, patient services and entitlements, statutory policy text. |
| **lib/analytics/facilityMetrics.ts** | Derived measures — facility scorecards, travel-time savings, mortality-prevention telemetry. All computed, none asserted. |
| **stores/** | Nine Zustand stores: patient, queue, referral, appointment, facility, resource, directory, auth, language. |
| **components/gov/** | Shared government-portal chrome. `PortalShell`, `GovPanel`, `Breadcrumb`, `PortalAside`, header and footer. See section 7. |
| **public/sw.js** | The service worker. Bump `CACHE_VERSION` whenever you change the pre-cache list, or clients keep the stale shell. |
| **scripts/** | 37 verify suites (`npm run verify`) and three browser checks (`check:a11y`, `check:offline`, `check:production`). CI runs them all. |
| **server/** | The mesh relay: `mesh-server.ts` on a server, `relay/vercel.ts` hosted; rules in `relay/app.ts`. |
| **backend/** | The district record service (FastAPI). See backend/README.md. |

## Conventions you'll notice

- **Trilingual by inline object.** Most pages define a local `txt` or `pageTexts` object with `en / hi / mr` keys. `lib/i18n.ts` exists but the inline pattern is dominant — follow the file you're editing.
- **Verification scripts use dynamic imports.** `lib/` is CommonJS, so the `.mts` suites use `await import()`, not static imports.
- **Dates resolve after mount.** This is a static export: computing "today" during render bakes the build date into the HTML and then mismatches on hydration. Use `useEffect`.

# 4. The data layer

## Thirteen object stores (schema v6)

| Store | Holds |
|---|---|
| **patients** | Patient demographics, ABHA id, vitals, triage status, risk flags |
| **queue** | OPD tokens — registration, call and completion timestamps |
| **referrals** | Tier-to-tier transfers with status and dispatch timestamps |
| **appointments** | Citizen bookings that convert into queue tokens |
| **diagnostics** | Test orders and results |
| **medicineStock** | Facility-level stock against the IPHS essential drug list |
| **facilities** | The seven facility records |
| **facilityResources** | Beds, equipment and their status, per facility |
| **maintenanceLog** | Equipment faults and repairs |
| **notifications** | Referral and system notices for this device's users |
| **users** | Staff accounts this device knows (its own facility's PIN hashes only) |
| **auditLog** | Every clinical change, with actor and timestamp |
| **syncQueue** | The outbox: records waiting to upload to the district service |

## Two rules enforced in lib/db.ts

**1. Writes throw; they never silently no-op.** If a referral cannot be found, the update raises rather than returning quietly — a silent failure in a clinical path is worse than a crash.

```
if (!ref) throw new DatabaseError(`Referral ${id} not found — status not updated to ${status}`);
```

**2. Audit logging never throws.** The inverse rule. If the audit write fails, the clinical action has *already* been applied — so it logs the error and carries on rather than rolling back real care.

## The audit trail

- Every change to a referral, queue entry, appointment or stock line records the acting staff member and a timestamp.
- The actor is module-level state — set it with `setCurrentActor(id, role)` on login.
- **Never write patient names into audit rows.** The audit surface is readable by the District Health Officer; it stores the token number and the record id, which already resolve to the patient for anyone entitled to see it. A PII leak here was found and fixed once already.
- Viewing is restricted to the `DHO` and the `SUPER_ADMIN`. The same screen shows the district service's access log: who read or uploaded which record, refused attempts included.

# 5. The triage engine — read before touching

This is the highest-risk code in the project. It decides how urgently a patient is seen.

## Three decision paths

| Path | What it is |
|---|---|
| **NEURAL_NETWORK** | A TensorFlow.js model scores the case from vitals and symptoms and returns RED, YELLOW or GREEN with a confidence. |
| **CLINICAL_OVERRIDE** | Deterministic danger-sign rules sitting *above* the model — altered consciousness, shock-range blood pressure, critical respiratory rate, severe hypoglycaemia and others. |
| **RULE_ENGINE** | The model could not run — still training, timed out, or failed on a low-end phone. The IPHS rules alone decide, and the slip says so. |

**The safety invariant — do not break this**

**Overrides escalate only.** A clinical rule can raise a patient's urgency. Nothing — not the model, not any later step in the flow — may lower a level that a rule has set. In a triage tool the failure that matters is the one that sends someone home. If you change this file, the escalate-only property must still hold.

## How it is guarded

`npm run verify:triage` runs 20 clinical cases through all three decision paths, the rules alone included. It must stay at 20/20. Add a case whenever you add a rule. These cases guard against regressions; they are not clinical validation. The model is trained on a synthetic dataset, and before real use the rules need a clinical review and the tool a validation study (docs/PRODUCTION_READINESS.md). The result slip states that it is decision support and the health worker decides.

### Priority levels

| Level | Meaning | Routing |
|---|---|---|
| **RED** | Emergency | Immediate care or transfer to the highest reachable tier |
| **YELLOW** | Urgent | Seen the same day; escalate if vitals move |
| **GREEN** | Routine | Standard OPD flow with follow-up scheduled |

# 6. The modules

Nine functional modules, all writing to the same local database. A patient's record travels to the facilities that need it — the one that registered them, any facility a referral of theirs touches, and the district officers — so a patient referred from a sub-centre is already known at the hospital that receives them, and nowhere else.

| Module | Route | What it does |
|---|---|---|
| **OPD Intake & Triage** | /opd | Registration, ABHA capture, vitals entry, runs triage, issues a token |
| **District Command** | /dashboard | Indicators, facility network status, high-risk action list |
| **OPD Queue** | /queue | Live token board; wait times computed from real call timestamps |
| **Referrals** | /referrals | Tier-to-tier transfer with live transit tracking |
| **Appointments** | /appointments | Citizen booking; slot capacity derived from actual facility staffing |
| **Medicine Stock** | /medicine | Stock register: counted receipts and issues, low stock, near expiry, requisition |
| **Diagnostics** | /diagnostics | Test orders and results against the patient record |
| **High-Risk Follow-up** | /followup | Scheduled recall for ANC, SAM and chronic cases |
| **Teleconsult** | /teleconsult | A structured record beside an eSanjeevani call. No audio or video of its own, by design. |
| **Pre-Arrival Board** | /incoming | Patients en route to this facility, from the district service |
| **Facilities & Services** | /facilities, /services-info | Directory, entitlements, scheme eligibility (PMJAY gated to CHC and above) |
| **Audit Trail** | /audit | Device trail and district access log; DHO and Super Admin only |
| **Administration** | /admin | Staff accounts and PINs; Super Admin only |

Six staff roles scope what each user sees and does: ANM, Medical Officer, Hospital Admin, Specialist, District Health Officer, Super Admin (`lib/auth/permissions.ts`, mirrored to the district service).

# 7. UI conventions

The interface deliberately imitates an Indian government department portal — NIC built, GIGW 3.0 compliant. It is modelled on india.gov.in, mohfw.gov.in and arogya.maharashtra.gov.in. This is a requirement, not a stylistic preference.

## Do

- Square corners. Radius 0–2px. Portals are built from bordered boxes.
- Separate content with a **1px rule**, never a drop shadow. All elevation is flattened globally in `tailwind.config.ts`.
- Put content in a `GovPanel` — a solid navy heading bar above a square bordered body.
- Present data in **tables** with a header row and alternating row shading.
- Keep type small and dense: body 12–13px, table text 12px, labels 11px.
- Use `PortalShell` for the three-column layout.

## Do not — these read as machine-generated

- Coloured left accent rails on cards (the rotating blue/orange/green stripe). These were removed from seven files; do not bring them back.
- Large KPI tiles with oversized display numbers. Use a bordered indicator table.
- Pill badges carrying a checkmark glyph, such as "Verified" or "Ready".
- Rounded cards with soft shadows in a uniform three-across grid.
- Icons inside coloured circles.
- Hover elevation.
- Emoji, anywhere.

**Layout trap that has already bitten once**

`Sidebar` is a fixed `w-64` with `flex-shrink-0`, because most pages still use the legacy flex layout. `PortalShell` therefore uses **explicit grid tracks** — `16rem / minmax(0,1fr) / 20rem` — not 12-column fractions. Changing the sidebar to `w-full` makes it swallow the whole row on every unmigrated page. One page (the command centre) uses PortalShell; the rest still use the flex layout.

# 8. The no-invented-data rule

The most important convention in this codebase, and the one most easily broken by accident.

It is tempting, when a demo looks empty, to pad a number so the screen looks populated. In a medical tool that is not a cosmetic choice — it puts a figure in front of a clinician that nothing backs. Every one of the patterns below was found in this codebase and removed.

| Anti-pattern | Why it is wrong | Do instead |
|---|---|---|
| `patients.length + 152` | Inflates a real count with an invented offset | Show the real count, even if it is 5 |
| `count > 0 ? count : '4'` | Falls back to a fake number when the truth is zero | Show zero — zero is information |
| `78% Occupied` **hardcoded** | Asserts a measurement nobody took | Compute from facility bed records |
| **"Avg response: 28 min"** | A plausible-looking figure with no source | Derive from timestamps, or omit |
| **Hardcoded dated notices** | Stale the day after they are written | Compute forward from today |

## What good looks like

- **Bed occupancy** sums `beds.total` and `beds.occupied` across facility records, and renders "Not reported" when no facility has recorded capacity.
- **Queue wait times** are computed from `calledAt - registeredAt`. Where nobody has been called yet, the board says so instead of showing a placeholder.
- **Announcements** list the next occurrence of each recurring clinic day, computed from `nextOccurrence()` — so the dates move on their own.
- **Referral transit clocks** only run where a real dispatch timestamp exists. A referral with no dispatch shows no clock rather than a fabricated elapsed time.

> **Rule of thumb**
>
> If you cannot point at the record a number came from, it does not go on the screen. State a real zero, or say "no data".

# 9. Running and verifying it

## Local development

```
git clone https://github.com/TharunS2006/HealthcareAI.git
cd HealthcareAI && git checkout new
npm ci
npm run dev # app on :3000
npm run dev:full # app on :3000 + mesh relay on :3001
```

## Before every commit

```
npm run lint # zero warnings
npm run typecheck
npm run verify # all 37 suites, with a summary
npm run build # static export, then stamps the service worker
npm run check:offline # opens the ANM's screens with the server stopped
npm run check:a11y # WCAG 2.1 A/AA on 28 pages
npm run check:production # the first day's flow, in production mode
```

## Testing offline properly

Do not trust DevTools' "Offline" toggle — during this project it left `navigator.onLine` true and requests still succeeded. **Kill the server instead** — `npm run check:offline` does exactly that. And check that screens *start*, not just render: an earlier worker cached each page's HTML but not its JavaScript, and screens looked right but did nothing. It passed manual tests only because the browser's own HTTP cache happened to hold the files.

## Gotchas that cost time

| Symptom | Cause and fix |
|---|---|
| **A route 404s in dev but is fine in the build** | Stale dev route manifest, usually after `npm run build` wiped `.next` under a running dev server. Stop dev, `rm -rf .next`, restart. |
| **Pages render unstyled** | Same cause — the build removed the dev server's assets. |
| **Offline start misses a new page** | You added a route without adding it to `PRECACHE_ROUTES` and bumping `CACHE_VERSION` in `public/sw.js`. |
| **Hydration mismatch on a date** | A date computed during render in a static export. Move it into `useEffect`. |
| **Sidebar swallows the page** | See the layout trap in section 7. |

# 10. Where the project stands

## Verified working

| Check | Status |
|---|---|
| **Production build** | Passes |
| **Lint and TypeScript** | Clean — zero warnings, no errors |
| **Verify suites** | 37 / 37, including the district service on PostgreSQL |
| **Triage clinical suite** | 20 / 20, through all three decision paths |
| **Offline** | Every ANM screen starts with the server stopped (check:offline) |
| **Accessibility** | No axe-core WCAG 2.1 A/AA violation on 28 pages (check:a11y) |
| **Production flow** | Accounts, a RED referral, the pre-arrival board, the access log, revocation and a relay restart, in a browser against real servers (check:production) |
| **Dependency audits** | npm audit and pip-audit: no known vulnerabilities |

## Known gaps — be honest about these

- **Not yet cleared for real patients.** The licence, clinical validation, the CDSCO view of triage, a CERT-In-empanelled audit, government hosting and ABDM certification are all still due — docs/PRODUCTION_READINESS.md lists each, with who owns it.
- **No restore to a replacement device.** Records reach the district service, but a new phone cannot download its facility's records back yet.
- **Teleconsult has no audio or video**, by design: it is a structured record beside an eSanjeevani call. Do not describe it as video consultation.
- **The UI rework is partial.** One page uses `PortalShell`; the rest still use the legacy flex layout. They have picked up the global token changes and lost their accent rails, but their card grids have not yet become tables.
- **The seven statutory policy pages are English-only** while the other citizen-facing pages are trilingual. Defensible — most state portals do the same — but inconsistent.
- **Several staff screens are English only** (referrals, pre-arrival board, ANM dashboard, patient record, administration); translation needs the department's language cell and a clinical review.

## Suggested first tasks for a new developer

- Read `lib/db.ts` end to end, then `lib/triage/model.ts`.
- Run `npm run verify` and read the suites for the area you will touch — they state what must hold.
- Migrate one legacy page to `PortalShell` — `/queue` is a good first one — and convert its card grid to a table.
- Do the offline test yourself, by killing the server. It is the clearest way to understand what the architecture actually buys.

> **If you remember three things**
>
> **1.** The device is the source of truth — the servers are couriers and oversight. **2.** Triage overrides escalate only. **3.** Never put a number on screen that you cannot trace to a record.